



NAFIRA

**AI METABOLIC
INTELLIGENCE**

METABOLISM OS

Aarush
Diwakar

1. Executive Summary

Nafira is a cutting-edge web application designed to revolutionize personal nutrition through Artificial Intelligence. By integrating advanced computer vision (Gemini Vision AI) and Large Language Models (Mistral-7B), Nafira transforms the traditional, manual process of calorie tracking into a seamless, data-driven experience. The platform aligns with United Nations Sustainable Development Goal 3 (Good Health and Well-being) by providing users with accessible, scientific clarity on their metabolic health.

2. Problem Statement

Current nutritional advice is often generic, disjointed, and difficult to adhere to. Users struggle with:

- **Manual Entry Fatigue:** tedious logging of ingredients and weights.
- **One-Size-Fits-All Advice:** Lack of personalization based on unique biological data.
- **Disconnect from Sustainability:** Little awareness of the environmental impact of food choices.

3. The Solution: Nafira

Nafira addresses these challenges by acting as a "metabolic operating system." It leverages AI to "see" food differently—analyzing meals physically and chemically to predict their impact on the user's specific biology.

Core Features

- **AI-Powered Meal Analysis:** Users simply upload a photo of their meal. The system, powered by **RTQVLM-0.5b & Gemini Vision AI**, identifies ingredients, estimates portions, and calculates comprehensive nutritional data (calories, macros) instantly.
- **Personalized Metabolic Insights:** Using **Mistral-7B**, the app generates contextual health advice tailored to the user's age, goals, allergies, and activity level.
- **Sustainable Food Dashboard:** Tracks the environmental footprint of dietary choices, encouraging eco-friendly eating habits.
- **Real-time Health Analytics:** A comprehensive dashboard visualizes calorie intake, macro distribution, and historical trends using interactive charts.

4. Technical Architecture

Nafira is built on a robust, modern full-stack architecture designed for performance, scalability, and security.

Frontend

- **Framework:** React 18 with React Router
- **Styling:** Tailwind CSS (custom design system)
- **Animation:** Framer Motion (premium UI feel)
- **Visualization:** Recharts (data analytics)

Backend & AI

- **Server:** Node.js & Express
- **Database:** SQLite (lightweight & fast)
- **Vision AI:** Google Gemini Vision API & RTQVLM-0.5b (if available locally)
- **LLM Engine:** Mistral-7B
- **Auth:** JWT & Bcrypt encryption

5. Sustainability & Ethics

Beyond personal health, Nafira emphasizes **Global Health**. The "Sustainability Dashboard" actively calculates the carbon footprint of meals, gamifying the reduction of environmental impact. This dual-focus approach promotes personal longevity alongside planetary sustainability.

6. Conclusion

Nafira bridges the gap between complex nutritional science and daily human behavior. By automating data entry and personalizing insights, it empowers users to make informed decisions that improve their metabolic health without the cognitive burden of traditional tracking methods. It is not just a calorie counter; it is an intelligent companion for a healthier, more sustainable life.

MADE WITH 11.3K LINES OF CODE
2026 NAFIRA, GitHub: [rusetiq/nafira](https://github.com/rusetiq/nafira)